

# MIRA mini

A Lightweight Non-Diagnostic Prototype for Observing AI–User  
Conversation Dynamics

Jun Sakai

Independent Researcher, Yokohama, Japan

ORCID: 0009-0006-3194-9990

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## Abstract

MIRA mini is a lightweight local prototype for observing short AI–user conversations as dynamic conversational fields. It is designed as a small “conversation air meter” rather than as a diagnostic, psychological, medical, or safety-scoring tool. Given a short conversation, it visualizes gentle indicators such as openness, loopiness, AI amplification, return path, and overall conversation air, while also exporting exploratory research metrics such as effective rank, compression index, seed echo ratio, seed-to-cue conversion, and cue amplification. MIRA mini deliberately avoids classifying users or judging conversations as safe or dangerous. It provides a local, non-persistent interface for noticing whether a short AI conversation appears open, repetitive, amplified, or able to retain a return path. The prototype is intended for personal exploration, educational use, and non-commercial research.

## 1 Introduction

AI systems are increasingly used not only as information tools, but also as conversation partners, writing assistants, learning aids, business assistants, and personal companions. As a result, human language is often shaped through interaction with AI before it enters wider social, institutional, or creative contexts. This creates a need for lightweight ways to observe AI–user conversational dynamics without turning such observation into diagnosis, surveillance, or risk labeling.

MIRA mini addresses this need as a small local prototype. It lets a user paste a short AI conversation and view several gentle indicators of the conversational field. The tool is intentionally modest: it is not a clinical, psychological, safety-critical, or dependency-detection system. Its purpose is to make conversational-field observation tangible and reproducible in a simple research prototype.

This framing is consistent with classic work on conversation as an organized sequence of turns, grounding, and mutual coordination. Turn-taking research emphasizes that conversation is structured through sequential exchange, while grounding theory describes how participants build and maintain shared understanding during interaction. Human–computer interaction research has also examined how longer-term relationships can emerge through repeated interaction with computational agents.

Recent work and public discussion around AI interpretability have emphasized the need to understand and observe increasingly capable AI systems. For example, Goodfire describes its work as interpretability research for understanding and intentionally designing advanced AI systems, while Anthropic frames interpretability as a foundation for AI safety and positive outcomes. MIRA mini differs from internal interpretability approaches: it does not inspect model weights, activations, or hidden states. Instead, it observes only the visible conversational surface between a user and an AI system. MIRA mini adopts this general conversational perspective in a lightweight computational prototype, without attempting to diagnose users, evaluate AI systems, or certify safety.

## 2 Conceptual Background

MIRA mini is part of the broader MIRA project, which explores AI–user conversations as dynamic fields rather than as isolated messages. In this framing, each user utterance and AI response can influence the direction, repetition, amplification, and openness of the next exchange. MIRA mini adopts only a minimal public-facing subset of this idea. It focuses on lightweight indicators that can be shown gently in a local interface, without diagnosing users, classifying conversations, or determining safety.

## 3 Prototype Design

MIRA mini is implemented as a local Streamlit application. The user pastes a short AI conversation, typically a few recent exchanges. The application then computes a small set of conversational indicators and displays them using a soft visual interface.

The design principles are:

1. **Local use:** the prototype is intended to run on the user’s local machine.
2. **No registration:** no login, account, or user registration is required.
3. **No persistent storage:** conversation logs are not designed to be stored persistently.
4. **Non-diagnostic framing:** outputs are not diagnoses, scores, warnings, or safety judgments.
5. **Gentle presentation:** visible labels use ordinary language such as openness, loopiness, return path, and conversation air.
6. **Research transparency:** detailed metrics can be exported as CSV for non-commercial research and reproducibility.

## 4 Observed Indicators

MIRA mini presents five primary user-facing indicators.

### 4.1 Openness

Openness represents whether the conversation appears to retain semantic or directional breadth. In the user interface, this is presented gently as a tendency of the conversation field, not as a judgment of quality or safety.

## 4.2 Loopiness

Loopiness represents whether the conversation appears repetitive, circular, or narrowly self-reinforcing. The term is intentionally softer than “fixation” or “risk” so that the prototype remains non-diagnostic.

## 4.3 AI Amplification

AI amplification represents whether the AI response appears to intensify, echo, or enlarge cues from the user’s input. In MIRA mini, this is treated only as a lightweight observation, not as evidence of risk, dependency, or unsafe interaction.

## 4.4 Return Path

Return path represents whether the conversation appears to retain a route back to broader, grounded, or non-escalating directions. It does not determine whether a conversation is safe. It only indicates whether the current field appears to preserve some movement away from narrowing or over-amplification.

## 4.5 Conversation Air

Conversation air is a compact overall presentation layer. It summarizes the current atmosphere in gentle language. It is intended to communicate that MIRA mini is an air meter, not a diagnostic instrument.

# 5 Research Metrics

For research use, MIRA mini can export additional metrics. These are intended to support exploratory analysis and should not be interpreted as direct psychological or safety measurements.

| Metric                 | Interpretation in MIRA mini                                                                     |
|------------------------|-------------------------------------------------------------------------------------------------|
| Effective rank         | Approximate dimensional breadth or diversity of the conversation representation.                |
| Compression index      | Degree to which the conversation representation appears compressed or narrowed.                 |
| Seed echo ratio        | Degree to which early user cues are echoed or preserved in later AI response patterns.          |
| Seed-to-cue conversion | Degree to which initial cues become amplified or transformed into stronger conversational cues. |
| Cue amplification      | Overall tendency for cues to be enlarged or intensified across the interaction.                 |

Table 1: Research-oriented metrics exported by MIRA mini. These metrics are exploratory indicators, not diagnoses or safety scores.

## 6 Implementation

The public prototype consists of a minimal Streamlit interface and a separate engine module. A typical public repository structure is:

```
app_en.py
mira_engine_en.py
requirements.txt
README.md
CHANGELOG.md
LICENSE
NOTICE.md
assets/
```

The application can be run locally with:

```
pip install -r requirements.txt
streamlit run app_en.py
```

The interface uses a simple visual design with soft colors and a cloud motif. This choice is not merely decorative: it supports the framing of the tool as a conversation-air meter rather than a judging or warning system.

## 7 Non-Diagnostic Boundary

MIRA mini is explicitly not:

- a diagnostic tool;
- a psychological assessment tool;
- a medical or mental-health evaluation tool;
- an AI-dependency detector;
- a safety classifier;
- a risk-scoring system;
- a tool for judging users, conversations, or AI systems.

The prototype must not be used to diagnose individuals, evaluate mental health, classify users as dependent or unsafe, monitor people without consent, rank, evaluate, or certify AI systems, or provide medical, psychological, legal, or safety-critical judgments. The intended uses are personal exploration, educational use, non-commercial research, and prototyping of conversational-field observation methods.

## 8 Limitations

MIRA mini has several limitations. First, the indicators are exploratory and prototype-level. They should not be treated as validated psychological, clinical, or safety measures. Second, short conversations may not provide enough data to support stable interpretation. Third, the visible labels are intentionally simplified for usability and should not be confused with formal measurement categories. Fourth, any embedding-based or rule-based method is sensitive to language, wording, model choice, and preprocessing. Fifth, MIRA mini observes only pasted text and does not access internal model states. These limitations are part of the design boundary. MIRA mini is intended to make a small research idea visible, not to make authoritative claims about users or AI systems.

## 9 Conclusion

MIRA mini provides a small, local, non-diagnostic prototype for observing AI–user conversation dynamics. The central contribution is not a validated safety classifier, but a framing and implementation pattern: AI–user conversations can be observed as dynamic fields without turning users into diagnostic objects. By keeping the tool local, non-persistent, non-commercial, and explicitly non-diagnostic, MIRA mini offers a modest first step toward accessible conversational-field observation.

## References

- [1] OpenAI. The next phase of enterprise AI. OpenAI, 2026.
- [2] Goodfire. Goodfire AI: Interpretability research for understanding and intentionally designing advanced AI systems. Goodfire, 2026.
- [3] Anthropic. Interpretability research. Anthropic Research, accessed 2026.
- [4] Harvey Sacks, Emanuel A. Schegloff, and Gail Jefferson. A simplest systematics for the organization of turn-taking for conversation. *Language*, 50(4):696–735, 1974. doi:10.2307/412243.
- [5] Herbert H. Clark and Susan E. Brennan. Grounding in communication. In L. B. Resnick, J. M. Levine, and S. D. Teasley, editors, *Perspectives on Socially Shared Cognition*, pages 127–149. American Psychological Association, 1991. doi:10.1037/10096-006.
- [6] Timothy W. Bickmore and Rosalind W. Picard. Establishing and maintaining long-term human-computer relationships. *ACM Transactions on Computer-Human Interaction*, 12(2):293–327, 2005. doi:10.1145/1067860.1067867.

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## Competing Interests

The author declares no competing interests.

## Data and Code Availability

MIRA mini is intended as a local, non-persistent prototype. No user conversation logs are stored by the tool. The source code, sample inputs, and example outputs will be made available through the public project repository and archived with the Zenodo record.